

Abductive Inference¹

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¹Slides 1-10 are based on the paper “Logic-Based Abductive Inference” by Sheila McIlraith
<http://www.cs.toronto.edu/~sheila/publications/abduction.pdf>

Goal of Abduction

- You have a propositional logic theory Σ about the world. (Can be easily adapted to FOL as well).
- You have a set O of observations.
- If $\Sigma \models \bigwedge_{o \in O} o$, then clearly Σ fully explains all the observations in O .
- But what if $\Sigma \not\models \bigwedge_{o \in O} o$?
- In this case, Σ does not fully capture what is happening in the world.
- Abduction tries to come up with *explanations* of the observations O that are as close as possible to Σ .

Abductive Framework

Abductive Framework

An abductive framework is a pair $(\Sigma, \mathcal{E})$ such that:

- Σ is a finite set of propositional formulas.
- $\mathcal{E}$ is a finite set of propositional formulas called *abducibles*.

Example

$\Sigma =:$

$$m_1 \leftarrow d_1$$

$$m_2 \leftarrow d_1$$

$$m_1 \leftarrow d_2$$

$$\mathcal{E} = \{d_1, d_2, d_3\}.$$

More Examples

MORE EXAMPLES ON THE BOARD

Abductive Explanation

Abductive Explanation

Suppose $(\Sigma, \mathcal{E})$ is an abduction framework and O is a finite set of observations. $E \subseteq \mathcal{E}$ is an *explanation* iff:

- ① $\Sigma \cup E \models \bigwedge_{o \in O} o$, and
- ② $\Sigma \cup E$ is unsatisfiable.

E is a *minimal explanation* of O iff E is an explanation of O and there is no strict subset $E' \subset E$ that is also an explanation of O .

Example

Example

$\Sigma =:$

$$m_1 \leftarrow d_1$$

$$m_2 \leftarrow d_1$$

$$m_1 \leftarrow d_2$$

$\mathcal{E} = \{d_1, d_2, d_3\}$.

Suppose $O = \{m_1, m_2\}$. Then $E_1 = \{d_1\}$ and $E_2 = \{d_1, d_2\}$ are both explanations of O — but E_1 is the only minimal explanation.

Abductive Explanation as a Set Cover Problem

- Peng and Reggia looked at a set D of diseases and a set M of manifestations.
- They assumed a *causation* relation $C \subseteq D \times M$. $(d, m) \in C$ is read as “disease d causes manifestation m . Many different diseases may cause the same manifestation.
- Given a set $M^+ \subseteq M$ of observed manifestations, they define a cover of M^+ to be a set $D_I \subseteq D$ such that $\{m : | (\exists d \in D_I)(d, m) \in C\} \supseteq M^+$. That is, every manifestation that is observed is covered by at least one disease in D_I .
- D_I is a *parsimonious cover* of M^+ iff D_I is a cover of M^+ and no strict subset of D_I is a cover of M^+ .

Because the causation relation can be encoded as a definite clause, the previous definition of abductive explanation subsumes this definition.

Consistency-Based Hypothesis

Consistency-based Hypothesis

Suppose $(\Sigma, \mathcal{E})$ is an abductive framework and O is a set of observations. An explanation $E \subseteq \mathcal{E}$ is a consistency based hypothesis iff $\Sigma \cup E \cup O$ is consistent.

- 1 Every abductive explanation is a CBH.
- 2 Vice-versa not necessarily true.

Example

Example

$\Sigma =:$

$$m_1 \leftarrow d_1$$

$$m_2 \leftarrow d_1$$

$$m_1 \leftarrow d_2$$

$\mathcal{E} = \{d_1, d_2, d_3\}.$

- $\emptyset$ is a CBH.
- $\{d_1, d_3\}$ is a CBH.

Least Presumptive Abductive Explanation

LPAE

Suppose $(\Sigma, \mathcal{E})$ is an abduction framework and $D_1, D_2 \subseteq \mathcal{E}$ are explanations. E_1 is **less presumptive than** E_2 iff $\Sigma \cup D_2 \models D_1$. In this case, D_2 is **more specific than** D_1 .

Disjunctive Logic Programs

Disjunctive Logic Program (DLP)

If $A, B_1, \dots, B_n$ are atoms, then:

$$A_1 \vee \dots \vee A_m \leftarrow B_1 \& \dots \& B_n$$

is a disjunctive rule. A disjunctive logic program (DLP) is a finite set of disjunctive rules.

An Herbrand interpretation I satisfies the above rule (assuming it is ground) if either:

- $(\exists i \in [1, m]) A_i \in I$ or
- $(\exists j \in [1, n]) B_j \notin I$.

This definition of satisfaction is extended to non-ground rules and DLPs in the obvious way.

Minimal Models of Disjunctive Logic Programs

Minimal Models Disjunctive Logic Program

An interpretation I is a minimal model of a DLP P iff:

- I is a model of P and
- there is no $I' \subset I$ s.t. I' is a model of P .

Closed World Assumption

Consider the DLP P given by just one rule:

$$a \vee b \leftarrow$$

- Using CWA, see that we can infer both $\neg a$ and $\neg b$ from P .
- But this seems inconsistent with P .

EXAMPLES ON BOARD

Generalized Closed World Assumption

Generalized Closed World Assumption (GCWA)

Suppose P is a DLP and a is a ground atom.

$P \models_{GCWA} \neg a$ iff for all minimal models M of P , $M \not\models a$.

In our example with $P = \{a \vee b \leftarrow\}$, we now see that $P \not\models_{GCWA} \neg a$ and likewise $P \not\models_{GCWA} \neg a$.

Transforming Abduction to Minimal Models

We can transform the Peng-Reggia causation relation $C \subseteq D \times M$ into a disjunctive LP P as follows.

- 1 For all manifestations m , insert the rule

$$\bigvee_{(d,m) \in C} d \leftarrow m$$

into P .

- 2 If $o \in O$ is an observation, then insert the fact

$$o \leftarrow$$

into P .

Minimal models correspond to explanations.